System Architecture & Design Document: Multi-Agent Research Factory

Technical Specification of 5-Agent Coordination, Pub/Sub Bus, and Epistemic Feedback Loops

Course: F0003 CA2/CA3 · Agentic AI Course
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Repository: https://github.com/AnushkaTandon26/Multi-Agent-Reseach-and-Report-Factory

1. Architectural Overview & Orchestration Pattern

The platform implements a hybrid Orchestrator-Evaluator-Optimizer architecture. A central ResearchOrchestrator manages system state and coordinates agent lifecycles over an asynchronous Event Bus. Rather than executing a rigid cascade, the orchestrator facilitates bidirectional negotiation between specialized worker agents and the adversarial Critic. Each agent is modeled with distinct system directives, specialized tool suites, and memory isolation.

2. Agent Roles and Core Responsibilities

• Agent 1: Strategic Planner (PlannerAgent) — Analyzes user thesis, determines target audience, defines hypotheses, and outputs structured research vectors.
• Agent 2: Literature Scout (SearcherAgent) — Operates WebSearchTool and AcademicSearchTool, querying global search APIs, arXiv, and encyclopedic repositories.
• Agent 3: Deep Reader (ReaderAgent) — Implements DocumentChunkingTool and FactExtractionTool to distill high-density factual claims.
• Agent 4: Epistemic Critic (CriticAgent) — Audits facts against hallucination matrices, executes adversarial red-team stress tests, and triggers ReplanDirectives.
• Agent 5: Executive Synthesizer (WriterAgent) — Authors structured whitepapers with executive summaries, comparative tables, LaTeX math, and podcast briefing scripts.

3. Inter-Agent Communication & Pub/Sub Message Bus

All agents communicate via a typed, audited in-memory Message Bus. Every message carries a unique ID, sender agent ID, recipient agent ID, ISO timestamp, message type, payload, and causal parentMessageId. This architecture guarantees full auditability, deterministic execution tracing, and enables real-time visual inspection in the Web UI.

4. The Non-Linear Critic Feedback Loop

If CriticAgent review produces an overallCredibilityScore < 85/100 or discovers unsupported claims, it emits an AGENT_MESSAGE with type "critic_rejection". The orchestrator increments the replanLoop counter, updates the Planner with targeted gaps, and re-dispatches the Scout to retrieve corroborating references. Once satisfied (or max loops reached), synthesis proceeds to the Writer.

5. Technology Stack & Framework Justification

• Frontend: Next.js 14 App Router, React 18, Tailwind CSS, Lucide Icons with custom Liquid Glass, Claymorphism, Synthwave, and Cyberpunk design tokens.
• Multi-Agent Core: TypeScript, custom asynchronous pub/sub event bus, isolated tool harnesses.
• Model Integration: Google Gemini 1.5 Flash / Pro via @google/genai with fallback simulation engine.
• Testing & Verification: Vitest automated test suite with 6 test files covering tool execution, message bus, orchestrator, and feedback loop mechanics.